

Testing the system resources: In this testing technique, aspects like memory usage, auto deletion of temporary files, and issues linked to the growth of the local database are tested. In the process, the battery or power consumption issues are also analysed by the test engineers.

Performance testing: This tests the performance or efficiency of the mobile app by transforming the network connection from 2G or 3G to Wi-Fi, sharing documents, gauging battery consumption and other aspects. The mobile app is tested under different scenarios so that its behaviour can be analysed under certain conditions such as low battery, bad network coverage, low available memory, simultaneous access to the application's server by several users, and other conditions. The performance of an application can be affected by what's happening on the server side and client side. So performance testing is carried out to check both.

Operational testing: This is done to test backups and recovery plans when the battery goes down. Or if data gets lost while upgrading to a new application bought from the store.

Security testing: These tests are done to validate whether the mobile app is fully secured against cross site scripting, SQL injection, session hijacking, cookie stealing and other related attacks.

Functional testing: Functional testing makes sure that the application is working as per requirements. Many of the tests conducted for this are driven by the user interface and call flows.

Laboratory testing: Laboratory testing is usually done by network carriers, by simulating the entire wireless network. This test is performed to find out any glitches when a mobile application uses voice and/or data connections to perform some functions.

Testing for memory leakages: Memory leakage occurs when a program or mobile app is unable to manage the memory allocated to it, resulting in poor performance of the app and the overall slowing down of the system. As mobile devices have significant constraints on available memory, memory leakage testing is important for the proper functioning of any application.

Interrupt testing: Mobile apps face different interruptions like incoming calls, and problems related to battery power or network coverage. Interrupt testing ensures the avoidance of these issues.

Different types of mobile phone interrupts are: incoming and outgoing SMS and MMS, incoming and outgoing calls, incoming notifications, battery removal, cable insertion and removal for data transfer, network outage and recovery, media player being turned on or off, and the device's power cycle.

Installation testing: Many mobile apps come pre-installed on the mobile phone, whereas others have to be installed from the store. Installation testing ensures that the installation process goes smoothly without any problems for the user. This testing process covers installation, updation and uninstallation of the application.

Use of emulators

Currently, almost every organisation makes use of emulators as well as simulators in its research and development cells. Using such technologies, the look and feel of the device or application can be experienced without using the actual infrastructure. With this method, there is no damage or maintenance of actual infrastructure, because everything is tested virtually. The use of emulators is extremely useful in the initial stages of development, as they allow quick and efficient checking of the app. An emulator is a system that runs software from one environment in another environment without changing the software itself. It imitates the features and behaviour of the real or actual system to be used.

Some mobile emulators are listed below.

- **Device emulator:** Packaged and provided by device manufacturers
- **Browser emulator:** Used to simulate mobile browser environments
- **Operating systems emulator:** Apple provides emulators for iPhones, Microsoft for Windows phones, and Google for Android phones

Challenges in mobile app testing

Mobile app testing can be challenging because of the following factors.

The sheer variety of mobile devices: Mobile devices differ in screen sizes, input methods (QWERTY, touch or normal) and have different hardware capabilities.

Diverse mobile platforms/OSs: There are many different mobile operating systems in the market. The major ones are Android, iOS, BREW, BREWMP, Symbian, Windows Phone and BlackBerry (RIM). Each operating system has its own limitations. Testing a single application across multiple devices running on the same platform and then on every platform, poses a unique challenge for testers.

Mobile network operators: There are over 400 mobile network operators in the world, of which some are CDMA and

Online mobile device emulators and testing tools

Using mobile device emulators, any Web application can be tested on different platforms or mobile devices. The following websites provide mobile device emulators.

<https://appthwack.com/>
<https://testobject.com/>
<http://www.testiphone.com/>
<http://iphone4simulator.com/>
<http://iphone-emulator.org/>
<http://iphone5csimulator.com/>
<http://www.mobilephoneemulator.com/>
<http://ready.mobi>
<http://responsivepx.com/>
<http://quirktools.com/screenfly/>
<http://ipadpeek.com>
<http://transmog.net/iphone-simulator/>